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1. Executive Summary & Problem Statement

ETIOMS is a full-stack web application designed to empower Ethiopian companies in discovering, qualifying, and tracking procurement tenders. By allowing companies to register detailed business profiles (sectors, regions, experience, certifications), the system leverages AI to score tenders against these profiles, conduct preliminary compliance audits, answer contextual questions from tender text, and manage the complete bid lifecycle.

1.1 Problem Statement

- **Fragmentation:** Tenders are scattered across newspapers, disparate web portals, and physical notice boards, resulting in no centralized search repository.
- **Complex Compliance:** Tender documents are extensive and laden with strict eligibility rules (e.g., trade licenses, tax clearance, VAT/TIN registration, bid security). Missing a single mandatory document results in immediate disqualification, often after non-refundable bid bonds have been paid.
- **Manual Tracking:** Deadlines and opportunity pipelines are tracked manually, increasing the risk of missed submissions.

1.2 System Capabilities

Capability	Summary
Tender Ingestion	Upload a PDF/TXT or paste a notice. Gemini extracts title, entity, sector, region, deadline, budget, requirements, and generates a structured summary.

Capability	Summary
Search and Filtering	API filters by sector, region, status, keyword, and days-until-deadline. Dashboard provides keyword, sector, and minimum-score filters.
Match Scoring	0–100 score per tender based on the user's profile: 70 points rule-based + 30 points AI semantic analysis, with a per-factor breakdown.
Compliance Audit	Cross-references the company’s certifications against mandatory documents, returning *BID*, *NO-BID*, or *CONDITIONAL* with disqualification risks.
Tender Q&A	Natural-language inquiries are answered exclusively from the tender’s text, featuring citations or an explicit "not available" safeguard.
Opportunity Pipeline	Tracks bids across seven stages: Discovered, Interested, Analyzing, Preparing, Submitted, Awarded, Not Awarded.
Comparison	Facilitates side-by-side analytical comparison of 2 to 5 targeted tenders.
Alerts	In-app notifications for open tenders closing within a 7-day window.
Teams and Admin	Supports organizational roles, single-use 24-hour invitation codes, and an administrative console featuring system statistics.

> *Note: All AI output is explicitly labeled in the user interface as advisory and must be verified against official tender notices.*

2. System Architecture and Technology Stack

The ETIOMS platform operates on a decoupled client-server architecture utilizing Next.js for the frontend and FastAPI for the backend, supported by a relational database and Google Gemini for AI inference.

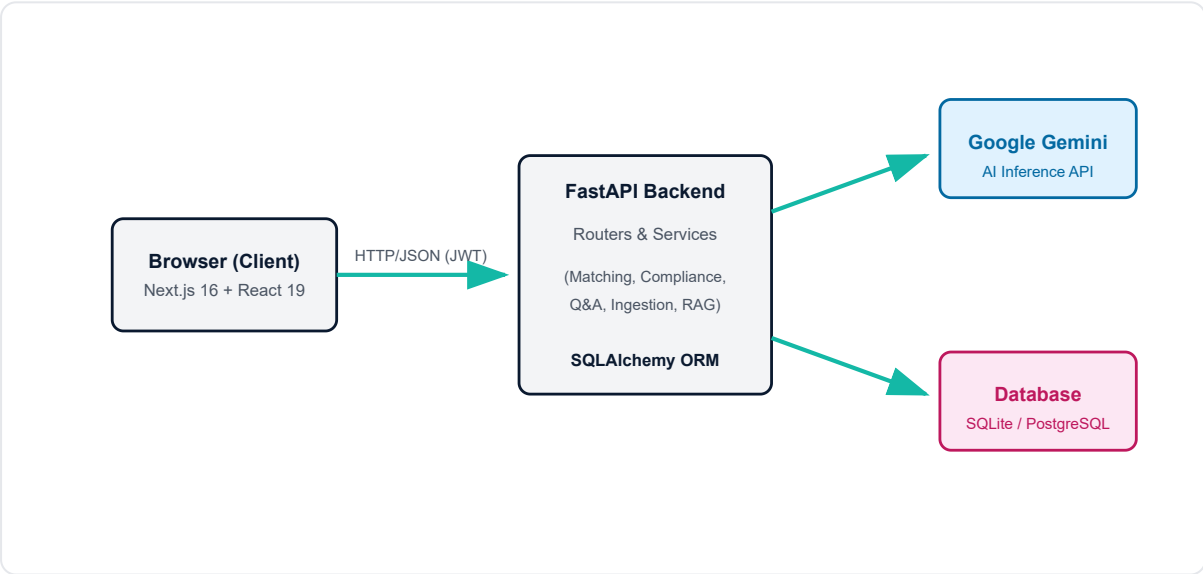


Figure 1: ETIOMS High-Level System Architecture

2.1 Technology Stack

Layer	Technology
Frontend	Next.js 16.4, React 19.3, TypeScript, styling in `globals.css`
Backend	Python, FastAPI, Uvicorn, Pydantic v2

Layer	Technology
Database	SQLAlchemy 2 ORM. SQLite by default, PostgreSQL by setting `DATABASE_URL`
Authentication	`python-jose` (JWT, HS256), `passlib` (PBKDF2-SHA256)
Documents	`pypdf` for PDF text extraction
AI Integration	Google Gemini (`google-genai` SDK): `gemini-3.6-flash` for extraction, scoring, audit, Q&A, and proposals; `text-embedding-004` for embeddings

3. Database Design

The system utilizes seven core tables created automatically at startup. Primary identifiers are UUIDs stored as `CHAR(36)`. Dependent rows are declared with `ON DELETE CASCADE`.

Table Name	Key Fields	Description
`organization_profiles`	`id, company_name, sectors (JSON), operating_regions (JSON), years_experience, certifications (JSON), past_projects_summary`	One row per company; serves as the baseline profile utilized for tender scoring.
`users`	`id, email (unique), user_name, hashed_password, organization_id, role, is_active`	Supports "Organization Admin" or "Team Member" roles.
`organization_invites`	`id, code_hash, organization_id, created_by_user_id, expires_at, used_at`	Stores a SHA-256 hash of single-use codes.
`tenders`	`id, title, procuring_entity, sector, region, deadline, status, description, budget, requirements (JSON), raw_document_path, ai_summary,`	A centralized, shared catalog; not exclusively owned by any single organization.

Table Name	Key Fields	Description
	extracted_requirements, embedding	
tender_matches	id, profile_id, tender_id, total_score, is_qualified, compliance_decision, ai_justification, breakdown (JSON)	Cache-aside mechanism storing match results to optimize AI inference costs.
opportunity_tracking	id, organization_id, tender_id, status, notes, updated_by_user_id, created_at, updated_at	Records for the pipeline workflow.
notifications	id, user_id, type, message, read_flag, created_at	System alerts specific to individual users.

4. Core System Implementation & AI Workflows

4.1 Tender Ingestion Pipeline

Tenders are ingested via file uploads (PDF/TXT) or raw text. For PDFs, `pypdf` extracts text while injecting `--- Page N ---` markers for citation accuracy. The first 30,000 characters are transmitted to Gemini with a strict schema to extract metadata. PDFs without a text layer are rejected (no OCR).

4.2 AI-Powered Tender Matching Workflow

The evaluation engine calculates a 100-point compatibility score, requiring 60 points for a "qualified" (BID) status.

Component	Points	Evaluation Rule
Sector	25	Case-insensitive substring match between company and tender sectors.
Region	20	Matches if company lists 'Ethiopia' or the specific tender region.
Experience	25	Calculated as `years / 3 * 25` (capped at 25 points).
AI Semantic Fit	30	Gemini LLM rates certifications, past projects, and experience against the tender requirements and summary.

4.3 Compliance Audit

The system executes three strict rule-based checks: trade/business license, tax clearance, and VAT/TIN. Gemini, prompted as an Ethiopian public-procurement (FPPA) auditor, then returns a decision (BID, NO-BID, or CONDITIONAL), a justification, disqualification risks, and safeguards to complete before sealing the bid envelope.

4.4 Tender Q&A

Up to 40,000 characters of source text are sent to Gemini with instructions to answer only from that text and cite sections or pages. The reply explicitly handles ungrounded requests by answering ‘This information is not available in the provided tender documents’ when the text does not contain the answer.

5. API Reference

The backend exposes 29 endpoints. All endpoints require `Authorization: Bearer <token>` except those designated as public.

Group	Method and Path	Purpose

System/Auth	`GET /health`
	`POST /api/auth/register`
	`POST /api/auth/token`
		Health check (public).
		Create company account or join by invitation.
		Log in, receive JWT.

Tenders	`GET /api/tenders`
	`POST /api/tenders/upload`
	`GET /api/tenders/semantic-search`
	`POST /api/tenders/{id}/ask`
		List with filters (sector, region, status, deadline).
		Upload PDF/TXT or paste text and ingest via AI.
		Rank tenders by vector embedding similarity.
		Query a specific tender document.

Profile	`GET /api/profile/match/{id}`
	`GET /api/profile/compliance/{id}`
	`GET /api/profile/matches`
		Retrieve match score (cached; supports force_refresh).
		Execute FPPA compliance audit.
		All tenders ranked by score (min_score filter).

Group	Method and Path	Purpose
Pipeline	`GET /api/opportunities` `PUT /api/opportunities/{id}/status`	List the company's tracked opportunities. Update stage and tracking notes.

6. Frontend Architecture

The frontend uses the Next.js App Router. The JWT is kept in `sessionStorage`; a 401 response clears it and returns the user to sign-in. The API address comes from `NEXT_PUBLIC_API_BASE_URL`.

- **Ranked Opportunities (`/`):** Metric cards, search, sector and minimum-score filters; tenders ranked by score; save to pipeline; select up to 5 to compare.
- **Tender Detail Drawer:** Score breakdown, AI justification, FPPA compliance audit, question box with citations.
- **Opportunity Pipeline (`/opportunities`):** Board of stage columns, stage dropdown on each card, notes.
- **Upload Tender (`/ingest`):** PDF/TXT upload or pasted text; shows extracted fields.
- **Company Profile (`/profile`):** Edit sectors, regions, experience, certifications, past projects.

7. System Implementation & Setup

```
 git clone -b finish-mvp https://github.com/bereketcodes/ETIOMS.git
cd ETIOMS
python -m venv venv
venv\Scripts\activate  # (Linux/macOS: source venv/bin/activate)
pip install -r requirements.txt
copy .env.example .env

# Start Backend
python -m uvicorn backend.app.main:app --reload --host 127.0.0.1 --port 8

# Start Frontend
cd frontend
```

```
npm install
npm run dev
```

The bundled database contains a demo company, AfroTech Solutions PLC.



8. Testing and Evaluation

The backend was exercised with a scripted API smoke test against a clean temporary database to verify fallback behaviors when the AI was unavailable.

Test Scenario	Result / Verification
Protected endpoint accessed without token	401 Unauthorized
Wrong password at login / Duplicate email	401 Unauthorized / 409 Conflict
Match score with a fully matching profile, AI unavailable	70 (25 sector + 20 region + 25 experience); message explains the AI part is unavailable.
Q&A without an API key	Returns a not-grounded message, no crash.
Tenant isolation	A second company sees an empty pipeline and gets 404 when updating the first company’s opportunity.

9. Limitations and Future Work

Component Area	Current Limitation	Proposed Future Work
Q&A Retrieval	The first 40,000 characters of the document are sent as one prompt. Citations are written by the model and not machine-checked.	Split documents into chunks, embed and retrieve the best chunks, and verify citations against page markers.
Authorization	Tenders are a shared catalogue. Admin check is looser than strict role comparison.	Add strict role dependency and restrict tender creation and invitations to admins.
Score Cache	Cached match scores are not cleared when the company profile changes.	Invalidate cached matches on profile update.
Ingestion / OCR	Scanned PDFs without a text layer are rejected. Unparseable deadlines become today's date.	Add OCR, flag uncertain fields for review, and generate unique file names.